

Silver ion exchange in glasses of the system $\text{Na}_2\text{O}/\text{Al}_2\text{O}_3/\text{B}_2\text{O}_3/\text{SiO}_2$

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Abstract

Glasses in the system $\text{Na}_2\text{O}/\text{Al}_2\text{O}_3/\text{B}_2\text{O}_3/\text{SiO}_2$ were studied with respect to their ionic conductivity and their silver/sodium interdiffusion coefficients. The densities and refractive indices as well as the optical transmission of the glasses before and after silver/sodium ion exchange were determined. Only trace quantities of silver occurred in the metallic state and the samples are widely transparent. With respect to the ionic conductivity, the glasses exhibit Arrhenian behavior. The specific ionic conductivities are correlated with the silver interdiffusion coefficients. The gradient in the refractive index increases with increasing Na_2O -concentration of the glass the gradient materials are prepared from, and decrease with increasing equimolar substitution of SiO_2 by B_2O_3 . Optimum glass compositions for the preparation of gradient index lenses possess B_2O_3 -concentrations in the range of 12.5–15 mol% and molar Al_2O_3 -concentrations few mol% larger than the respective Na_2O -concentration.

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1. Introduction

Gradient index optical glasses enable the preparation of lenses with two planar surfaces. Up to now, gradient index lenses are predominantly produced by melting thallium containing glasses and subsequent ion exchange in alkali containing salt melts [1,2]. The as produced lenses contain large concentrations of thallium. Since thallium compounds are extremely hazardous, great efforts are made to replace thallium containing glasses by environmentally safe compositions. To produce a gradient index lens by ion exchange, the compound diffusing in the glass should cause large changes in the refractive index and furthermore should exhibit large interdiffusion coefficients. Here, the most advanta-

geous non-hazardous compound is silver. For that purpose, glasses in the system $\text{Na}_2\text{O}/\text{Al}_2\text{O}_3/\text{B}_2\text{O}_3/\text{SiO}_2$ are already industrially used [3–8]. The alkaline concentrations have to be fairly large (>20 mol% Na_2O) in order to enable large differences in the refractive index and fast diffusion, i.e. comparably short production times. The lenses are produced as follows: first a glass rod with a diameter of up to 3 mm is prepared. Then a first ion exchange is carried out in an AgNO_3 -containing salt melt, until a uniform Ag_2O -concentration in the whole volume is obtained. In this state, around 90 % of the Na_2O is replaced by Ag_2O . In a second ion exchange step, the Ag_2O is exchanged against Na_2O ; this process is interrupted, if the required profile in the refractive index is achieved [9–13]. This profile is a parabolic one, in the middle, the Ag_2O concentration, and the refractive index is maximum while at the surface, the Ag_2O -concentration is close to zero and the refractive index, minimum. One drawback of this production procedure is the tendency of silver ions to be reduced at elevated

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temperatures [14,15]. This may cause a yellow to black coloration, depending on the concentration of metallic silver in the glass. It has already been shown that this effect is comparably small if the glass does not contain non-bridging oxygen [16,17]. For this purpose, the above mentioned system, $\text{Na}_2\text{O}/\text{Al}_2\text{O}_3/\text{B}_2\text{O}_3/\text{SiO}_2$ glasses with $[\text{Na}_2\text{O}] < [\text{Al}_2\text{O}_3] + [\text{B}_2\text{O}_3]$ have been developed [18]. One of the problems associated with these glasses is the relatively large crystal growth velocity, especially in the temperature range from 800 to 900 °C which is important with respect to the drawing of the glass rods [19]. For some applications, the differences in the refractive indices (outer part/middle of the lenses) should be larger than in the presently produced lenses. This can only be achieved by changing the glass composition, especially by larger Na_2O -concentrations.

This paper provides a study on the effect of the chemical composition on the silver interdiffusion coefficients, the refractive index, the optical transmission and the electric conductivity of glasses in the system $\text{Na}_2\text{O}/\text{Al}_2\text{O}_3/\text{B}_2\text{O}_3/\text{SiO}_2$.

2. Experimental procedure

The glasses were prepared from reagent grades $\text{B}(\text{OH})_3$, $\text{Al}(\text{OH})_3$, Na_2CO_3 and SiO_2 (quartz) in 300 g batches. They were melted in platinum crucibles at a maximum temperature of 1500 °C in a furnace with MoSi_2 heating elements. The melts were casted on a graphite mould preheated to 500 °C, given to a muffle furnace with a temperature of 600 °C and then allowed to cool to room temperature. The sample compositions are summarized in Table 1.

The ion exchange was performed in a melt with the composition $\text{AgNO}_3 \cdot \text{NaNO}_3$ at 400 °C. A part of the samples were kept for 12 h in this melt, they were used to determine the silver interdiffusion coefficient. Other samples were kept for 16 days; they were used to record transmission spectra, to determine refractive indices and

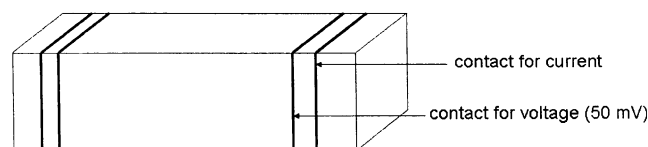


Fig. 1. Experimental set up for impedance spectroscopy.

densities. These samples are, in the following, denoted as ‘equilibrium ion exchanged’.

The densities were measured with a helium pycnometer (accupyc 1330, micromeritics). Refractive indices were measured at a wavelength of 546.06 nm (n_e) using a Pulfrich refractometer (PRZ, Zeiss, Jena, Germany). The glass transformation temperature of the as melted glasses was measured dilatometrically (dilatometer Netzsch 402 ES, Netzsch, Germany), using a heating rate of 5 K/min. Absorption spectra were recorded using a Shimadzu 3101 PC spectrometer. The silver interdiffusion coefficients were calculated from the profile of the refractive index measured by refracted near field technique at 632 nm [20]. Hereby, the refractive index was assumed to depend on the Ag_2O -concentration according to $n^2 = A[\text{Ag}_2\text{O}] + B$ (A and B = constants). It should be noted that the validity of this equation was proved in Ref. [21]. The electric conductivities were measured from glass samples with the geometry $4 \times 5 \times 70 \text{ mm}^3$ using the experimental setup shown in Fig. 1. The samples were contacted at four points, the outer two were used to measure the current while the two inner contacts were used to adjust the voltage to 50 mV. This was done potentiostatically using an impedance analyzer Zahner, Im5d, Zahner Elektrik, Kronach, Germany. The impedance spectra were measured in a frequency range from 1 to 10^6 s^{-1} at temperatures in the range from 200 to 600 °C.

3. Results

The impedance spectra recorded could be fitted within the limits of error to an equivalent circuit composed by a resistivity in parallel to a capacitance. In the frequency range from 10 to 10^3 s^{-1} , the phase angle was close to zero. The values of the total impedance in this frequency range is hence equal to the Ohmic resistance of the sample. Fig. 2 shows specific conductivities of the samples A, B, H and L as a function of the temperature as an Arrhenius plot. For all four glass compositions an approximately linear correlation between $\ln \sigma$ and $1/T$ is observed. From this correlation, the activation energy, E_a , was calculated. Specific conductivities at 400, 600 °C and at the glass transition temperature as well as the activation energies are summarized in Table 2. In Fig. 3, the effect of the B_2O_3 -concentration on the specific conductivity at the glass transition tempera-

Table 1
Chemical composition of the samples studied

Sample	$[\text{Na}_2\text{O}]$ in mol%	$[\text{Al}_2\text{O}_3]$ in mol%	$[\text{B}_2\text{O}_3]$ in mol%	$[\text{SiO}_2]$ in mol%
A	20	22.5	12.5	45
B	25	25	12.5	37.5
C	25	25	15	35
D	25	25	17.5	32.5
E	25	25	20	30
F	25	27.5	12.5	35
G	30	30	10	30
H	30	30	12.5	27.5
I	30	30	15	25
K	30	30	20	20
L	30	32.5	15	22.5

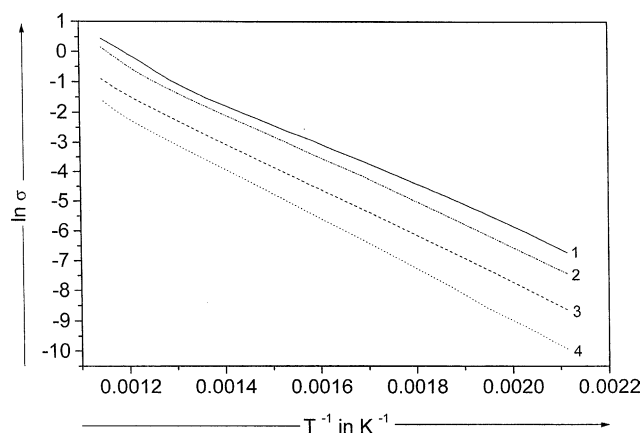


Fig. 2. Arrhenius plot of the specific conductivity of samples B (curve 1), H (curve 2), L (curve 3) and A (curve 4).

Table 2

Glass transition temperature, T_g , specific ionic conductivities at 400, 600 °C and T_g , and the attributed activation energy, E_a

Sample	T_g in °C	σ (400 °C) in S m^{-1}	σ (600 °C) in S m^{-1}	σ (T_g) in S m^{-1}	E_a in eV
A	569	0.008	0.199	0.058	0.78
B	569	0.100	1.574	1.053	0.577
C	535	0.015	0.401	0.121	0.73
D	526	0.012	0.432	0.093	0.737
E	510	0.008	0.38	0.058	0.78
F	594	0.055	0.858	0.778	0.666
G	600	0.070	1.232	1.232	0.63
H	567	0.069	1.177	0.694	0.637
I	561	0.032	0.572	0.31	0.667
K	520	0.014	0.496	0.098	0.76
L	574	0.025	0.411	0.237	0.876
Error	$\pm 5\text{K}$	$\pm 5\%$	$\pm 5\%$	$\pm 5\%$	± 0.01

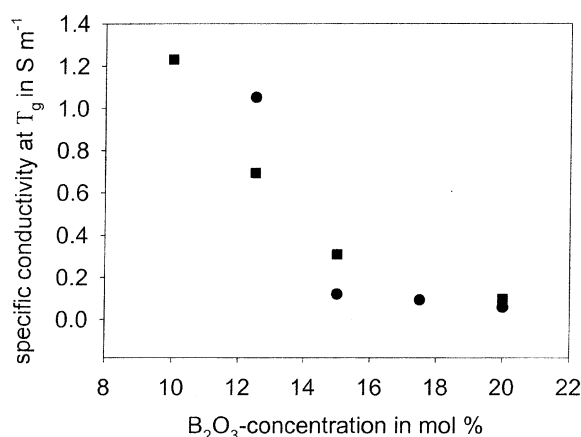


Fig. 3. Specific electric conductivity at the glass transformation temperature as an effect of the boron concentration for samples with $[\text{Na}_2\text{O}] = [\text{Al}_2\text{O}_3]$. (●) 25 mol% Na_2O and (■) 30 mol% Na_2O .

ture is shown for glasses with $[\text{Na}_2\text{O}] = [\text{Al}_2\text{O}_3]$. It is seen, that the conductivities decrease with increasing B_2O_3 -concentration. This effect is well pronounced and

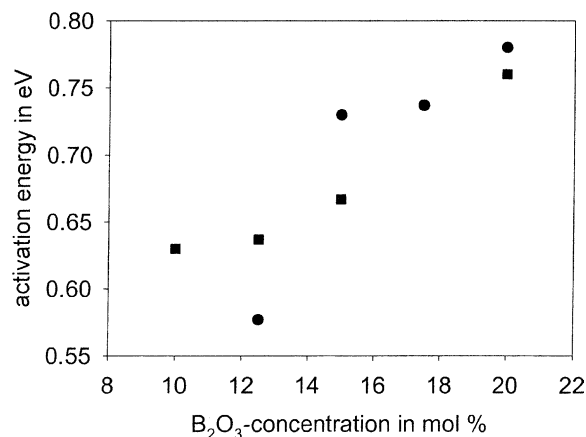


Fig. 4. Activation energy of the electric conductivity as an effect of the boron concentration for samples with $[\text{Na}_2\text{O}] = [\text{Al}_2\text{O}_3]$. (●) 25 mol% Na_2O and (■) 30 mol% Na_2O .

exceeds one order of magnitude. Fig. 4 presents the attributed activation energies of glasses with $[\text{Na}_2\text{O}] = [\text{Al}_2\text{O}_3]$. A general trend to increasing activation energies with increasing boron oxide concentrations is observed.

In Fig. 5, a typical diffusion profile is shown. It was obtained from sample B and is illustrated by the square of the refractive index. In the middle of the sample, the refractive index is constant and equal to that of the as melted glass. At the surface, the square of the refractive index is around 2.7, corresponding to $n = 1.64$. Here, the silver concentration is largest. In the first 0.2 mm, the diffusion profile fits well to an error function as shown by the dashed line. In the range from 0.2 to 0.4 mm, the values attributed to the error function are notably larger than the experimental values. This proves that the diffusion coefficient changes with the silver concentration in course of the ion exchange process. In the following, only the silver interdiffusion coefficient for

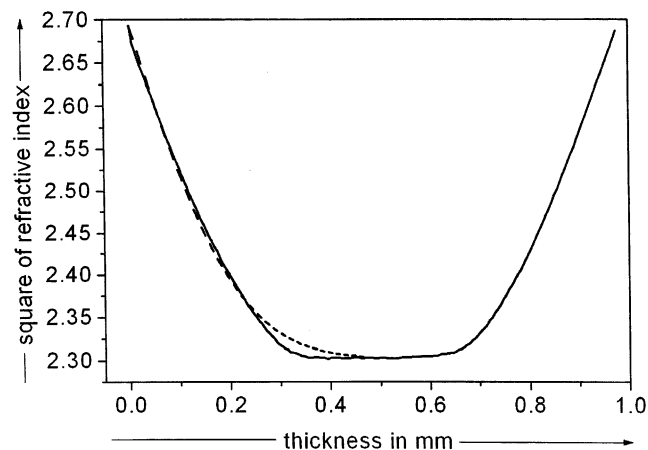


Fig. 5. Refractive index profile of sample B after ion exchange in a AgNO_3 containing melt (12h at 400 °C).

Table 3

Silver interdiffusion coefficient, D , density ρ of the as melted glass, density ρ_{Ag} of the ion exchanged glass, the refractive index, n_{Ag} , of the ion exchanged glass, the difference in refractive indices, Δn , (ion exchanged glass), the wavelength attributed to a transmission of 50%, λ_{50} , the transmission at a wavelength of 400nm, T_{400} , both of samples with standard geometry (1 mm thick)

Sample	D at 400°C in $10^{-9} \text{ cm}^2 \text{ s}^{-1}$	ρ in g cm^{-3}	ρ_{Ag} in g cm^{-3}	n_{Ag}	Δn	λ_{50} in nm	T_{400} in %
A	1.41	2.4083	3.1296	1.5908	0.0737	382	69
B	4.42	2.4437	—	1.6486	0.1264	373	79
C	2.028	2.418	3.495	1.6303	0.1154	387	63
D	1.815	2.4064	3.475	1.6271	0.1044	369	78
E	1.079	2.3926	3.3944	1.6179	0.1002	368	78
F	4.38	2.4508	3.6596	1.6450	0.1232	381	69
G	—	2.48	4.0331	—	—	417	24
H	6.016	2.4632	4.0403	1.6748	0.1519	415	25
I	4.56	2.4505	3.8156	1.6727	0.1492	408	41
K	1.667	2.411	3.6202	1.6581	0.1358	397	56
L	3.982	2.4632	3.797	1.6705	0.1448	403	46
Error	$\pm 5\%$	± 0.001	± 0.01	± 0.0001	± 0.0002	± 1	± 1

large silver concentration will be discussed. In Table 3, refractive indices and diffusion coefficients for all compositions studied are summarized.

In Fig. 6, the refractive index after equilibrium ion exchange for compositions with $[\text{Na}_2\text{O}] = [\text{Al}_2\text{O}_3]$ is shown as an effect of the B_2O_3 -concentration. The refractive indices decrease steadily with increasing B_2O_3 -concentration and are larger for the samples with 30mol% Na_2O than for those with 25mol% Na_2O . Fig. 7 presents the differences in the refractive indices of ion exchanged samples. In analogy to Fig. 6, also these differences decrease with increasing B_2O_3 -concentrations. While the differences are in between 0.1 and 0.126 for 25mol% Na_2O , they are in the range from 0.136 to 0.152 for 30mol% Na_2O . In Fig. 8, silver interdiffusion coefficients at 400°C are presented as a function of the boron oxide concentration for glasses with $[\text{Na}_2\text{O}] = [\text{Al}_2\text{O}_3]$. The diffusion coefficients decrease with increasing boron oxide concentration. At the same boron oxide concentration, they are larger in the glasses with 30mol% Na_2O than in those with 25mol% Na_2O .

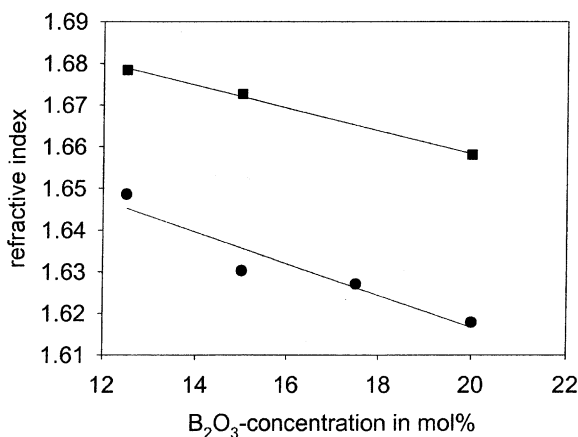


Fig. 6. Refractive index of totally ion exchanged samples $[\text{Na}_2\text{O}] = [\text{Al}_2\text{O}_3]$ as an effect of the boron oxide concentration (●) 25mol% Na_2O and (■) 30mol% Na_2O . Lines: regression lines.

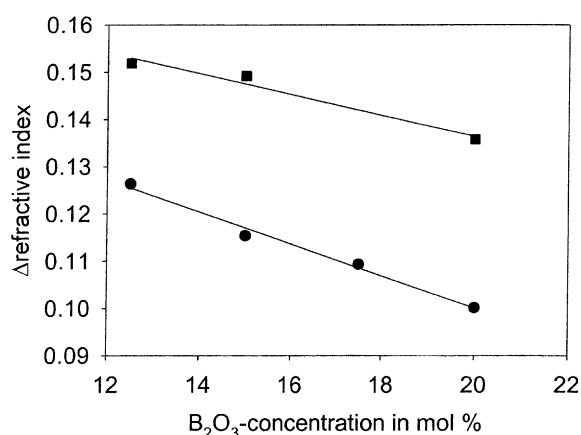


Fig. 7. Difference of the refractive indices of double ion exchanged samples ($n_{\text{inner}} - n_{\text{outer}}$) as a function of the total boron oxide concentration. (●) 25mol% Na_2O and (■) 30mol% Na_2O . Lines: regression lines.

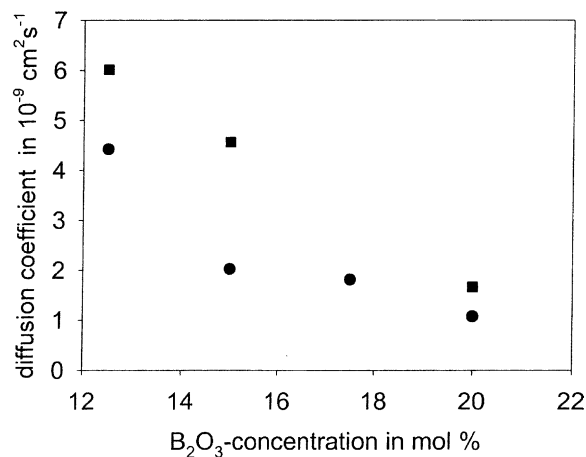


Fig. 8. Silver interdiffusion coefficient at 400°C as a function of the boron oxide concentration for samples with $[\text{Na}_2\text{O}] = [\text{Al}_2\text{O}_3]$. (●) 25mol% Na_2O and (■) 30mol% Na_2O .

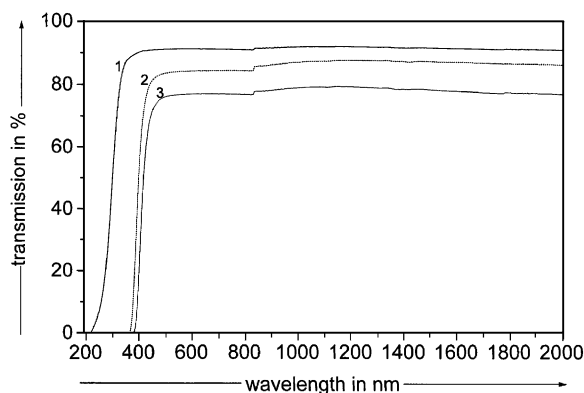


Fig. 9. Transmission spectra of glass samples (thickness: 1 mm). (1) sample K, (2) sample K after ion exchange and (3) sample G after ion exchange.

Fig. 9 shows optical transmission spectra of an as melted glass (sample K) and two glasses (samples K and G) after ion exchange. The UV-absorption edge in the as melted glass is around 300 nm, while it is shifted to the visible range after ion exchange. The effect can be illustrated by the wavelength, λ_{50} , attributed to a transmission of 50%. While this wavelength is around 300 nm in the melted glass sample K (approximately the same wavelength is observed in any glass sample studied), in the ion exchanged samples K and G, wavelengths of 397 and 417 nm are respectively observed. In Table 3, the values of λ_{50} and also the transmission at 400 nm are shown for all samples studied. The wavelengths λ_{50} are smaller for glasses with 25 mol% than for those with 30 mol% Na_2O . While the effect of the boron oxide concentration is small for glasses with 25 mol% Na_2O , the λ_{50} values of glasses with 30 mol% Na_2O decrease significantly with increasing B_2O_3 -concentration.

4. Discussion

All studied glass compositions, in principle, are suitable for an ion exchange in a silver containing salt melt. Ionic exchanged samples are widely transparent in the visible wavelength range. This proves that only trace amounts of silver occur in the metallic state as clusters. The shift of the UV-absorption edge after ion exchange is more pronounced in the glasses with 30 mol% Na_2O than in those with 25 mol% Na_2O . This is most probably due to the larger concentration of silver which occurs in the glass after ion exchange. The effect of the $[\text{Na}_2\text{O}]/[\text{Al}_2\text{O}_3]$ -ratio has also been shown in voltammetric studies [22] of glass melts in the system $\text{Na}_2\text{O}/\text{Al}_2\text{O}_3/\text{SiO}_2$. Here, depending on the $[\text{Na}_2\text{O}]/[\text{Al}_2\text{O}_3]$ -ratio, even a shift of the redox equilibrium with decreasing temperature to the metallic state, Ag^0 , has been reported. Since the ionic conductivity of the studied glasses is an effect

of the Na^+ -mobility, the effects of silver diffusivity and ionic conductivity will be discussed together. As shown in Fig. 3, the ionic conductivity strongly depends on the B_2O_3 -concentration of the glasses. Surprisingly, the variation of the conductivity with the B_2O_3 -concentrations is larger than the variation with Na_2O -concentration. The attributed activation energy increases with increasing B_2O_3 -concentration (see Fig. 4). Both indicate that the Na^+ -ions are more strongly bonded to the glass network if the B_2O_3 -concentration is larger. The mobility of the glass network, however, increases with increasing B_2O_3 -concentration as illustrated in Table 2, column 2, by the decrease in the glass transformation temperature.

In analogy to the ion conductivity, caused by the mobility of the Na^+ -ions, also the interdiffusion coefficient of silver decreases with increasing B_2O_3 -concentration. At 400 °C, this effect is clearly smaller (maximum variation: factor 5.6) than that concerning the ionic conductivity (maximum variation: factor 8). The silver interdiffusion coefficients in glasses prepared from those with 30 mol% Na_2O are always larger than those prepared from glasses with 25 mol% Na_2O . In analogy to Na^+ , also Ag^+ is more tightly connected to the glass network if the B_2O_3 -concentration is larger. Fig. 10 presents a plot of the silver interdiffusion coefficient versus the specific conductivity in a double logarithmic plot. As general trend, an increase of the diffusion coefficient with increasing conductivity is seen. This trend is observed for Na_2O -concentrations of 25 and 30 mol% and for those samples which possess an alumina concentration larger than the Na_2O -concentration.

The refractive index of silver containing samples (ion exchanged to equilibrium) is first of all a function of the Na_2O -concentration of the glass the samples are prepared from, however, it is also a function of the

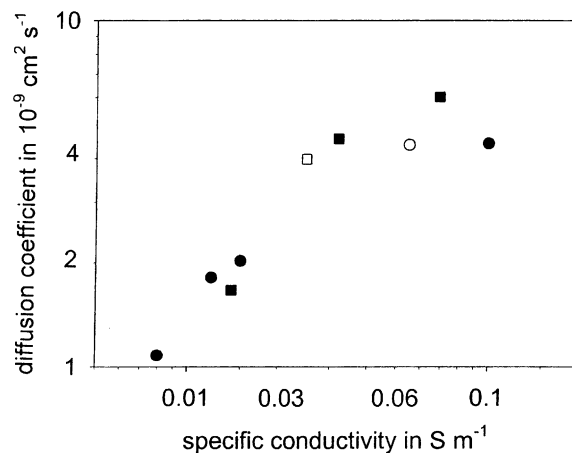


Fig. 10. Silver interdiffusion coefficient as a function of the specific electric conductivity at 400 °C. (O) 25 mol% Na_2O , ($\square$) 30 mol% Na_2O , full symbols: $[\text{Na}_2\text{O}] = [\text{Al}_2\text{O}_3]$, hollow symbols: $[\text{Al}_2\text{O}_3] = [\text{Na}_2\text{O}] + 2.5 \text{ mol\%}$.

B_2O_3 -concentration. Within the limits of error, a linear decrease of the refractive index with the B_2O_3 -concentration is observed. The attributed regression lines shown in Figs. 6 and 7 run approximately parallel and are shifted by around 0.03 to larger values in the glasses with 30 mol% Na_2O . The effect of the Na_2O -concentration is fairly clear, because during the ion exchange Na_2O is replaced (90%) by Ag_2O and therefore the concentration which leads to the increase in the refractive index is larger in the glass with the higher Na_2O -(or Ag_2O) concentration. The effect of the boron concentrations should first of all be seen in the smaller mean molecular weight of the respective glasses. If SiO_2 is substituted against an equimolar Al_2O_3 -concentration, the effect is qualitatively the same as in the case of B_2O_3 . For the glasses with 25 and 30 mol% Na_2O , the refractive index decreases by 0.0032 and 0.0044, respectively. In Fig. 11, the refractive index of ion exchanged glasses is shown as an effect of the respective density. The lines drawn stand for samples with 25 and 30 mol% Na_2O -concentration. For both Na_2O -concentrations, a steady increase of the refractive index with the density is observed. The samples with $[Al_2O_3] > [Na_2O]$ lie at the same line as those with $[Al_2O_3] = [Na_2O]$. The line attributed to the larger Na_2O -concentration is shifted to higher refractive indices, supposedly due to the higher concentration of Ag^+ in these glasses which is a highly polarizable ion.

For the preparation of gradient index lenses, not only the optical transmission and the gradient in the refractive index is of importance, but also the suitability of the glass for the rod drawing process, and the chemical durability. As recently reported [19], the crystal growth velocity of the as melted glass in the critical temperature range from 800 to 900 °C strongly depends on the chemical composition, and decreases if SiO_2 is substituted by

equimolar concentrations of B_2O_3 or Al_2O_3 . Otherwise, the chemical durability decreases, if SiO_2 is substituted by B_2O_3 and decreases with the Na_2O -concentration. An ideal glass for the preparation of gradient index lenses with differences in the refractive index of up to 0.125 should possess a Na_2O -concentration of 25 mol% and an Al_2O_3 -concentration few mol% larger (sample F). The latter suppresses the tendency to crystallization and leads to an increase in the chemical durability. As shown in Tables 2 and 3, the transmittivity at 400 nm, Δn and the silver interdiffusion coefficients are in the same range as the respective values of sample B. A further substitution of SiO_2 by B_2O_3 leads to increasing transmission at 400 nm and to smaller crystal growth velocities [19], however, it also results in smaller differences in the refractive index and in decreasing chemical durability [19].

In order to achieve larger differences in the refractive index ($\Delta n = 0.14 - 0.15$), the Na_2O -concentration of the glass has to be increased to 30 mol%. This leads to strongly increasing crystal growth velocities and drastically decreasing chemical durabilities. The equimolar substitution of SiO_2 by B_2O_3 decreases the crystal growth velocities, however leads to further decreasing chemical durabilities, and decreasing silver interdiffusion coefficients. In order to achieve crystal growth velocities smaller than $10 \mu m h^{-1}$ [19], the B_2O_3 -concentration should be around 15 mol% and, in analogy to the glass with 25 mol% Na_2O , the Al_2O_3 -concentration should be larger than 30 mol%. The resulting optical properties (see Table 3, sample I) are nearly as good as those of sample H (see Table 3) and the silver interdiffusion coefficient is nearly as large.

5. Conclusion

The specific ionic conductivities of the glasses exhibit Arrhenian behavior in the temperature range studied. The specific ionic conductivities as well as the silver/sodium interdiffusion coefficients decrease if increasing concentrations of SiO_2 are substituted against B_2O_3 . Ion exchanged samples are widely transparent in the visible range, the transmission at 400 nm decreases with increasing Na_2O -concentrations of the glasses the samples are prepared from, and decreasing B_2O_3 -concentration. The gradient in the refractive index which can be achieved by ion exchange and increases with the Na_2O -concentration of the glasses the samples are prepared from and decrease with increasing B_2O_3 -concentration. Optimum glass compositions to achieve differences in the refractive index in the range of 0.12–0.13 contain 25 mol% Na_2O and an Al_2O_3 -concentration few mol% larger. To produce differences in the refractive index in the range of 0.14–0.15, the Na_2O -concentration has to be increased to 30 mol% and also here, Al_2O_3 -

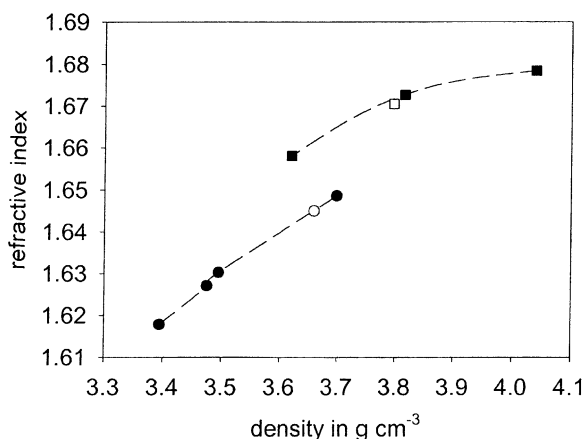


Fig. 11. Refractive index as a function of the density of totally ion exchanged samples. (O) 25 mol% Na_2O , ($\square$) 30 mol% Na_2O , full symbols: $[Al_2O_3] = [Na_2O]$, hollow symbols: $[Al_2O_3] = [Na_2O] + 2.5 \text{ mol\%}$.

concentrations few mol% larger are required. This is necessary in order to increase the chemical durability and decrease the crystal growth velocity.

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